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RESEARCH ARTICLE

IntPort: An Intelligent Portfolio Construction Technique Based on Financial Forecasting by Statistical Average Method

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ABSTRACT Investors are inclined to predict stock values, and conventional studies indicate that using machine learning (ML) for stock price prediction is the most effective method. This study focuses on ML-based stock price forecasting that aims to assist people with limited investment capabilities by providing a high-quality approach to price prediction. The research utilizes various characteristics of the current stock price, such as 'Open,' 'Low,' 'High, 'Adj Close', and 'Close'. In addition, it incorporates principal component analysis (PCA), linear discriminant analysis (LDA), and a statistical average method (SAM) to create additional features. These features and current stock price attributes are combined to form a feature vector for ML methods. The study also emphasizes building a portfolio that reduces investment risk while capitalizing on the expected upward movement of stock prices. Experimental results indicate that incorporating the SAM-price feature associated with the ML technique, bidirectional long short-term memory (Bi-LSTM), leads to superior forecasting accuracy compared to existing approaches. It is evident from previous literary works that various supervised learning methods have successfully predicted stock prices. Our research is dedicated to producing a novel approach to generate better price accuracy by forecasting and an effective portfolio that can be beneficial to the general public. The investigation reveals an accuracy rate of 98.14% when applied with the SAM-price feature defined by the additional feature dataset (AFD), surpassing the accuracy achieved by traditional pricing features alone. Additionally, as a novel price feature, AFD generates no over-fitting. Furthermore, the portfolio generated using the SAM-price function demonstrates a return on investment of 145.475% and a Sharpe ratio value of 77.62, indicating a favorable risk-return trade-off.

INDEX TERMS Stock price forecasting, machine learning, chi-square, PCA, feature extraction, LSTM.

I. INTRODUCTION

Presently, due to increased inflation majority of households are slopping to make at least 15% of their savings into various investments [1]. Investors with low contribution margins deal with various issues such as income, proper investment, stock selection, and background analysis to proceed with the price forecasting process and optimal portfolio performance. A "proper investment" refers to a situation in which a person can achieve the target return [2], [3] with respect to a given risk. Furthermore, asset selection determines the optimal allocation of stocks depending on the economic standing [4] market modality. Stock price forecasting refers to predicting stock prices for a specific period [5]. Besides, portfolio management is a way to optimize the performance of selected stocks created by the distribution of funds [6]. According to

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Andersson et al., it is difficult to get accurate information about the best investment decision to use stock features [1]. Hargreaves and Hao selected stocks using financial performance indicators [4]. In addition, from the perspective of price prediction, information about stock prices remains very noisy, stationary, and non-linear, so machine learning (ML)-based processes make better predictions according to Haq et al. [5]. Portfolio management ultimately allocates investment to reduce risk [5].

Machine learning methods are currently being used in research to anticipate stock price movements and stock selection [3]. Effective ML features for stock market prediction are developed from standard price features such as *Open, Low, High, Adj Close*, and *Close*.In our suggested study, feature values extracted using SAM (Statistical Average Method), LDA (Linear Discriminant Analysis), and PCA-based (Principal Component Analysis) approaches are compared against traditional price features to compare prediction performance. Higher accuracy and adjusted r-square indicate that the investment risk is mitigated and return is guaranteed. [4], [7]. As a result of the study, such a notion supports qualitative income development for lower-middle-class or middle-class investors who contribute small investments to the stock market. Using ML-based prediction, Haq et al. proposed that the return on investment (ROI) should be greater than the rate of return on the savings account's risk-free return [5]. Portfolio management compensates by recommending the optimum investment strategy for each individual based on income, budget, age, and risk tolerance by lowering risk while improving the possibility of return [8]. This proposes Sharpe ratio-based values for portfolio investment to generate a value greater than 1. Finally, we have concluded that, for predicting stock prices, Bi-LSTM remains the most effective method using the proposed price feature generated by the SAM method. Our proposed price dataset AFD derives a price accuracy of 98.14% using Bi-LSTM with an adjusted $r\hat{2}$ score greater than 99%.

A. RESEARCH OBJECTIVE & ARTICLE ORGANIZATION

This research is primarily intended to predict stock prices using novel price features. To accomplish this, we set the following objectives to meet our research goals and motivations:

- Selection of stocks that have high potential for future price growth through forecast analysis.
- Application of novel price features to make stock price predictions with better accuracy.
- Establishing SAM price as a novel perspective for future price prediction using ML-based price forecasts.
- Precisely indicating an optimized portfolio return using the AFD price dataset to achieve the research goals and introduce a novel approach to price forecasting.

To fulfill the research, this study accompanies 5 different sections of studies. To proceed with the research we have utilized chi-square test,[1] LIME,[2] and Shap[3] analysis to predict the novel features discriminatory and contributory aspects. Furthermore, we illustrated the usage of PCA[4] and LDA.[5] It will primarily introduce the research problem and how despite so much progress in ML-based forecasting, a novel approach has not yet been utilized in the price prediction method for stock prices. It will proceed with supplementing current research reviews and their gap analysis with a proposed overview and description of the experimentation process. In the experimental description, we will find the step-by-step procedure of the research method, and finally, in the analysis of the results, we will demonstrate the findings of the research. We will then conclude the research using the findings and suggest how this study has opened a new direction in the area of financial forecasting processes.

II. LITERATURE REVIEW

The ML technique uses stock price features as major predictor variables to predict prices [6]. Cakra and Trisedya pointed out that only one pricing feature is required to accurately anticipate stock prices [9]. Furthermore, Li et al. demonstrate that equities perform differently over time and with different financial parameters [10]. As a result, Mehtab et al. and Fisher and Krauss demonstrated that using several value features for prediction improves prediction accuracy [11], [12]. Furthermore, Hammoudeh et al. demonstrate that an appropriate portfolio for secondary income may be formed using supervised learning-based price estimation. Then, Zhu and Fan suggested that to measure the accuracy of portfolio performance for different portfolios various measures like the Sharpe Ratio or linear programming could be used [13]. Chen points out that supervised learning algorithms and classification-based regression can accurately predict stock prices [14], [15]. It's found that an effective contribution to reducing the attributes of PCA can help generate a quality feature [16]. Even though there are numerous works on ML-based stock price prediction, we have yet to find one that attempts to forecast the value of stocks using additive or derivative pricing characteristics. According to Shahi et al., the stock price feature is quite close in value, therefore applying a supplementary price attribute with the same or close-by levels does not improve the price projections adequately [17]. The movement of stock prices mainly depends on various market factors like beta or volatility, macroeconomic issues, financial conditions or ratios, news, and external factors [18]. Moreover, the significance of generated price features is highly applicable using both chi-square testing and contributory feature analysis. Hence

[1] https://scikit-learn.org/0.21/modules/sklearn.feature_selection.chi2.html
[2] https://christophm.github.io/interpretable-ml-book/lime.html
[3] https://towardsdatascience.com/using-shap-values-to-explain-how-your-machine-learning-model-works-732b3f40e137
[4] https://scikit-learn.org/stable/modules/sklearn.decomposition.PCA.html
[5] https://medium.com/@cxu24/common-methods-for-feature-extraction-pca-and-lda-7b1f5679e3bf

this research focused on modified price features to efficiently predict prices and generate profits.

Sumaiya & Aswani established the chi-square test's efficiency in identifying relevant features in ML-based estimation [19]. Cai et al. and Hussein and Özyurt demonstrated the importance of feature extraction for stock price prediction, implying that in several ML-based approaches, price prediction is carried out using one or more specific price features rather than all 5 price features [20], [21]. Zhong and Enke used principle component analysis (PCA) as an example of the mathematical benefit associated with predicting with just a single characteristic rather than numerous ones [22]. Long et al. claim that when all characteristic measurements are averaged, a new price attribute develops with a reduced standard deviation [23]. Consequently, in this research, we accumulated PCA-based features, linear discriminant analysis (LDA) based features, and mean value features by SAM (statistical averaging method) for price forecast using the ML method. Based on the chi-square test, if there is no significance in terms of a minimum confidence interval of 10%, then the feature is not good enough for prediction [24]. Li et al have shown that in the case of unsupervised cluster extraction, which is important for price feature selection, LDA (linear discriminant analysis) plays a pivotal role [25]. We have thus proposed a new price feature using LDA and the price feature development has been indulged in this research.

According to Lin et al., the primary financial metrics considered for stock selection in this study were earnings per share (EPS) and current ratio [26]. Both Amen and Goltz and Oh et al. demonstrated that, in the event of a competent forecasting system, the stock price may be reliably forecasted despite its extreme volatility [27]. Therefore, emphasis should be placed on the quality of the forecasting methods or process. As Haq et al. suggested ML methods currently predict stock prices with the highest accuracy [5].

According to Vadlamudi, stock price forecasting relies on ML techniques that use a set of criteria to produce qualitative forecasts [28]. Furthermore, Asad claims that supervised learning may efficiently predict stock price movement [29]. Because stock price data is erratic, Huang et al. recommend maintaining a selectable accuracy area for the forecast value range [30]. Huang et al. and Batra and Daudpota also suggested that in the case of margin of error allowance a range of at least 30% from the real value range could be considered a true positive [30], [31]. Hasnain et al. demonstrated that every parameter extracted from the confusion matrix has a significant influence in forecasting the expected result [32]. Gray et al. explain how, in the presence of noisy data, a variety of limitations can evaluate accuracy, precision, recall, and F1 score [33]. According to Deng et al., the quality parameters of the predicted value are evaluated by the classification report score defined by the confusion matrix [34]. According to Frías-Paredes et al., the mean absolute error (MAE) is more important than the root mean squared error (RMSE) for estimated values [35]. Willmott and Matsuura also demonstrate that MAE is more accurate in climate error prediction rather than RMSE [36]. Siew and Nordin demonstrated that the regression approach included sufficient data to justify MAE as an appropriate error computation method [37]. The anticipated price estimates shall be utilized in portfolio optimization after the prediction accuracy is evaluated using MAE.

Hyper-parameters are highly important for optimizing machine learning performance. It is seen that in various time series forecasting methods optimizing hyper-parameters plays a crucial role in determining optimized forecasting performance according to Sun et al. [38]. In this research, we have experimented with hyper-parameter tuning using each machine learning model and LSTM, Bi-LSTM, or GRU [39]. We have brought in various empirical evidence that supports the reasons for using various hyper-parameters for making the most effective performance in terms of predicting stock prices that will help make better portfolio optimization in case of additional gain. The ultimate target is to ensure the nonexistence of prediction over-fitting from the training and testing dataset.

Kao et al. have suggested that a classifier with increasing aptitude, epsilon between 0 to 0.1, and gamma values between 0 and 1 provides the best possible performance for support vector regression [40]. Similarly, Panwar et al. found that in a regression of the support vector nonlinear functions perform the best for price prediction or such dataset [41]. Hence a kernel value of radial basis function, polynomial, or sigmoid kernel performs the very best for training regression-related price datasets. Bao et al. suggested that in case price prediction where soft classification plays a crucial role the tolerance limit should be low [42]. In the case of Logistic regression Ahmed et al. have suggested that both the Solver algorithm of "lbfgs" and "sag" play a crucial role in nonlinear datasets [43]. Moreover, Arafa et al. found that a regularization strength of 1 means a 100% application would help make the best possible predictions. Finally, the absence of a penalty affects the prediction performance more accurately [44]. For tree-based regression processes, the prediction occurs by indulging the learning factors. But a good set of prior fixated hyper-parameters would always play a crucial role in prediction. It found that the maximum depth of tree models is important for all decision trees, random forests, and Xgboost-based regression [45], [46]. Roy et al. further suggested that the number of estimators should keep on increasing till the overfitting persists [47]. Hindrayani et al. further suggested that for both random forest and decision tree-based regressions the minimum leaf sample and minimum sample split should keep on increasing till overfitting occurs [48]. Finally using neural network-based supervised learning methods, we include GRU (gradient recurrent unit), LSTM (Long Short-Term Memory), and Bi-LSTM (Bi-directional Long Short-Term Memory) for price prediction [49]. Sunny et al. have suggested that in the case of neural networks, the initial learning rate should

vary between 0.0001 and 0.001 [50]. For GRU, Shah et al. supported the window size to vary between 1 to 30 [51]. Zhang et al. found the best possible price prediction accuracy using a density of 10 and a dropout rate of 20% using both LSTM and Bi-LSTM [52]. Shah et al. have also found that the activation function of "ReLU" performs the best in terms of stock price prediction and also suggested the number of nodes to be 10 for both LSTM and Bi-LSTM. Kabir et al. suggested that a decay rate of 0.97 and a momentum of 50% are highly effective for forecasting [53]. Finally, Hamayel and Owda suggested that in the case of non-linear predictions, it is better to provide an epoch of 50 and a batch size of 32 [54]. Hence for our prediction process we have thus engulfed the best possible hyper-parameters that have been induced by the Optuna that have produced most effective outcome [55]. Its shown that in time series prediction process Optuna plays a crucial role in understanding the most effective hyper-parameters which have thus made forecasting more effective and efficient [56].

In the context of portfolio-based optimization, PyPortfolioOpt, a Python-based optimization toolkit, uses factors like the Sharpe ratio and efficient frontier to optimize the investment returns process. According to Farinelli et al. and Bailey and Prado, the Sharpe ratio and efficient frontier are essential for portfolio profitability [57], [58]. According to Mancini et al. and Sadati and Nematian, the hypothesis of linear programming-based refinement gives optimal portfolio performance [59], [60]. Yudin's proposed PyPortfolioOpt library promotes optimal portfolio efficiency for the lowest amount of risk optimization [61]. Guerard et al.'s low-risk investment position for returns based on portfolio and the highest income has the most positive effect [62]. To understand the accuracy of forecasted prices we have followed a similar time-based classification problem addressed for promoting precision, recall, and F1 score of predicted prices. Hseih et al. have shown that for continuous time series processes, it's possible to generate a classification report by creating a limit of forecasted data [63]. The uniqueness of the research is to get a positive ROI by using assumptions defined by ML-based predictions and financial parameters to obtain a set of proposed values. The quality of the ML method is very important for effective evaluation. A portfolio is a collection of different investments in stocks. The investment proposition relies on the factors of making multiple less risky investments where return probability increases by Sharpe ratio value greater than 1 [64].

For stock price forecasts as well as priority data, ML approaches are proven [65]. Popular machine learning method considering a single price feature acquired from the "Adj Close" price feature, XGBoost can predict stock prices with a high degree of accuracy (71.23%) [66]. XGBoost, on the other hand, could forecast the value of stocks with an increased level of accuracy (79.12%) by leveraging "Close" as a real-time price feature [67]. Finally, where any random features are incorporated for stock price forecasting, XGBoost performs best (87.11%) [68]. Similarly, as multiple price variables are considered for forecasting in logistic regression (LogR), the prediction accuracy improves over just one price feature [69], [70]. For stock price forecasting using decision tree regression (DTR), just a single price feature was evaluated [48], [71]. However, when multiple price features are employed for predicting using random forest-based regression (RFR), accuracy improves [72], [73]. Furthermore, for stock price forecasting, the widely utilized support vector machine-based regression (SVR) which learns through classification by hyperplane has employed only one feature [40], [74], [75]. Finally, we discovered that the most popular stock price forecasting method, long short-term memory (LSTM), achieves greater accuracy with several price features [76], [77], [78]. But with "SAM Price" applied with conventional price features, we have found that Bi-LSTM makes the most accurate prediction using all the price variables. The results generated utilizing our portfolio and forecasted prices should outperform the S & P 500 and the best savings account return rates [79], [80]. In Table 1, we highlight issues using the opportunity gaps identified through various research. The factor of Sharpe ratio greater than one represents a higher return from an investment given the same level of risk [81] which remains our final parameter of performance measurement for both stock selection, performance of predicted stocks by both prediction accuracy and portfolio return accuracy. We found that in consequence, these researches didn't focus on the following issues.

- Statistically significant values were not obtained using the ML method.
- In most research, implied value is not employed for portfolio optimization.
- The significance of qualities in forecasting dependencies has not been examined.

Based on the accuracy of the prediction derived from much research, we can infer that Bi-LSTM is the most effective ML approach for price prediction. We also notice an increase in forecasting accuracy when stocks have low volatility and certain derived price attributes. Finally, this research focuses on prospective issues such as improved price forecasting and optimizing portfolio results that can deliver greater earnings for investors with limited investment capability.

A. CONTRIBUTIONS

The research sought efficient price forecasting to provide convincing investment benefits for investors whose contribution capability is small/meager. In this study, we explore the following scientific contributions to confirm this analysis:

- Extrapolating two more value features, where one is associated with the statistical average approach and the other for the PCA method and LDA approach.
- To determine the reliance of features generated using statistical average, PCA-based features methods, and

TABLE 1. Research findings and gaps.

Litterateur	Methods	Dataset Size	Findings	Gap Analysis
Y. Yang (2021)	XGBoost	7500	Feature Engineering generated better accuracy	Didn't experiment with Multivariable Features
Gumelar et al. (2020)	XGBoost	16665	'Close' Prices were used	Multivariable price predictions, hyper parameter tunings not updated
Gong & Sun. (2009)	LogR	21375	Logistic Regression parameters suited	Performed well with sample dataset but hyper parameter conditions ignored
Ali et al. (2018)	LogR	100280	Financial ratio-based asset selection with general regression condition	Tree-based optimal hyper parameters were not indulged in performance evaluation
Hindrayani et al. (2020)	DTR	4600	Fundamental price information is used for prediction	Optimized hyper parameter and multiple feature prediction is absent
Vijh et al. (2020)	RFR	10080	Used multivariable approach for prediction	Hyper parameter optimization and comparison condition is lacking
Henrique et al. (2018)	SVR	33661	Intensified boundary line training made prediction better	Didn't check the importance of hyper parameter tuning and feature optimization
Kao et al. (2013)	SVR	12300	Feature extraction helped generate better prediction	Hyper parameter optimization and multi feature based prediction ignored
Moghar & Hamiche (2020)	LSTM	2843	Single variable based LSTM training method applied	No hyper parameter optimization was adhered with best price feature choice
Wang et al. (2018)	LSTM	7250	Best price feature for price prediction is selected for training	Didn't check for better accuracy with multiple price features or optimized hyper parameters

LDA-based features by using the chi-square test, LIME, and Shap analysis.

- Analyze the accuracy of ML-based prediction algorithms for both volatile and nonvolatile stocks through standard deviation measurement by Sharpe Ratio.
- Choosing the best ML approach in terms of accuracy for both conventional statistical method, and augmented dataset-based prediction process.
- Using the predicted prices to generate and optimize a portfolio whose performance needs to be better than the investment standards of S & P 500 or savings account return.

III. PROPOSED METHODS

In Figure 1, this section describes the planned research approach for this topic. Asset selection, price data set compilation, price forecasting, and eventually portfolio optimization utilizing expected prices are all sequentially mentioned in this process.

The research method is explained below, step by step:

- Stocks are selected based on financial indicators proposed by Lin et al. using EPS, current ratio, and beta values [26]. Among the 42 sample stocks collected from *www.investing.com* for research purposes, we selected them with EPS being greater than zero, a current ratio value between one and two, and a beta of three different types, including high beta, low beta, and beta value of stocks close to one [82].
- Using principal component analysis, linear discriminant analysis, and statistical averaging methods, three additional price attributes were derived. "PCA compiled one-dimensional price feature", "LDA compiled one-dimensional price feature" and "Average Feature" are the names of these features. The derived features were combined with conventional price attributes in the research process to determine better price predictions. A chi-square test is performed on these price attributes to determine their significance for forecasting.
- All four price datasets such as "Price Based on Conventional Features", "Price Dataset by PCA Features" (including the PCA-derived price attribute), "Price Dataset by LDA Features" and "Average Feature-based price Dataset" (including the statistically average price attribute) are set up for chi-square testing, LIME, and SHAP analysis. As for the independence of the functions, the price attributes were tested. Any feature that fails the independence test by less than 10% is removed for stock price forecasting. The significance testing using chi-square is shown to exhibit LDA-based features incompetence.[6]
- Stock prices were predicted using ML methods such as logistic regression, support vector regression, decision tree regression, random forest regression, XGBoost regression, long-short-term memory, bidirectional long short-term memory, and Gradient Recurrent Unit. The predictions were performed using only multiple price features. The stock with the best ML prediction accuracy will be selected for portfolio building and optimization.
- Using a Python-based optimization library (PyPortfolioOpt[7]), we optimize stocks for lower risk and higher return potential. The derived portfolio is required to have a Sharpe ratio greater than one (1) and an ROI greater than the savings account return.

A. STOCK SELECTION

Stocks were selected based on financial performance such as EPS, current ratio, and volatility based on Lin et al.'s

[6]https://github.com/hsami180792/Test
[7]https://pypi.org/project/PyportfolioOpt/

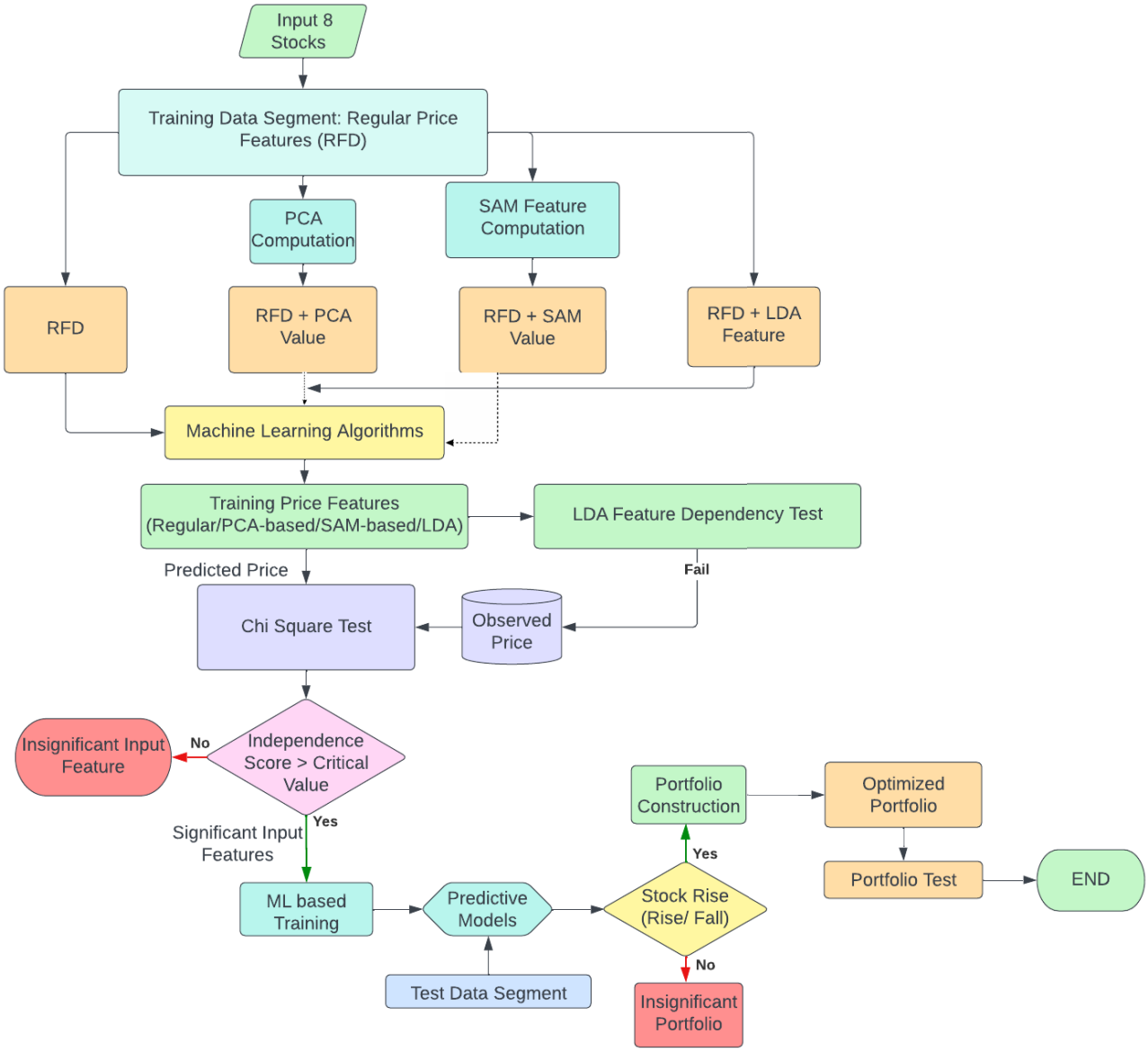

FIGURE 1. Proposed research model.

suggestion of financial parameters [26]. We considered the time of high fluctuation provided from mid-March 2020 till the end of June 2020 Pandemic COVID-19 to test the prediction accuracy of the ML-based techniques [83]. Stock selection based on financial ratios is illustrated in Figure 2.

B. DATA COLLECTION

Research corpus is found on the suggestive 42 stocks from investing.com,[8] investopedia.com,[9] and macrotrends.net[10] among which only 10 different stocks were selected for research purpose. Six of the selected stocks were found to have high and low volatility (beta greater than 1.5 or less than 0.5). This suggests that 60% of the stocks considered for the portfolio are volatile which can test the prediction quality purpose effectively. The remaining four have lower volatility (close to 1 of Beta value). Each stock has the same training and testing period. Total training data will include January 1, 2010, to March 15, 2020, a total of 2,783 days. This process is indulged by the initiation of main research-based forecasting. The testing and forecasting period has been underway from March 15, 2020, to June 30, 2020. The list of best-performing stocks is compiled. The research corpus is accumulated from *www.investors.com/research/best-stocks-to-buy-now/* and also from *www.investopedia.com/articles/stocks/06/ratios.asp*. For the

[8] https://www.investing.com/
[9] https://www.investopedia.com/
[10] https://www.macrotrends.net/

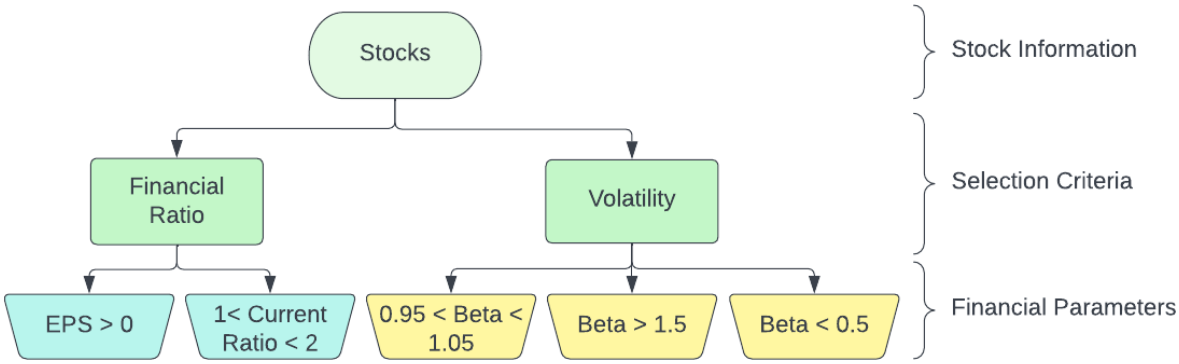

FIGURE 2. Stock selection criteria.

test of overfitting analysis we segregated the dataset into training, validation, and testing. We considered the period of 1st January 2020 to 15Th March 2020 to be the validation dataset. We performed the prediction on them to explain the overfitting analysis on the dataset. The results are exhibited in the Table 7.

1) FEATURE EXTRACTION

For this research, the common stock price attributes *'Open'*, *'High'*, *'Low'*, *'Close'*, and *'Adj Close'* are primarily tested using chi-square tests. This research uses PCA which reduces the dimension of input data by projecting it into a 1-dimensional price feature [53], and the resultant component is supported as an additional feature. This paper uses PCA to merge the measurements of five distinct pricing features into a single price feature, which is subsequently introduced as a new price. We have also introduced an additional price feature involving LDA based price feature. Furthermore, a new statistical averaging method (SAM) is utilized to create an extra attribute from the current stock price features. Moreover, a chi-square test is applied to determine the statistical significance of these features. Our method also accompanies the statistical significance of current stock price features in this research.

In the experimentation process, we have proposed price feature development using the statistical average technique, principal component analysis, and linear discriminant analysis. Our proposed research has generated new prices using LDA and these have been used for forecasting.

The method used for the extraction of features in addition to the data processing for chi-square testing is depicted in the figure 3. In our initial tests, we applied machine learning with feature vectors K1, K2, K3, K4, and K5, where K1, K2, K3, K4, and K5 represent five conventional attributes. Then, a new property, K(PCA), is formed from K1, K2, K3, K4, and K5. K(PCA) is joined with the previous feature vectors K1, K2, K3, K4, and K5 to generate new feature vectors $< K1, K2, K3, K4, K5, K(PCA) >$. Here, K(PCA) is also known as *'PCA-Price'*. The total properties are described in

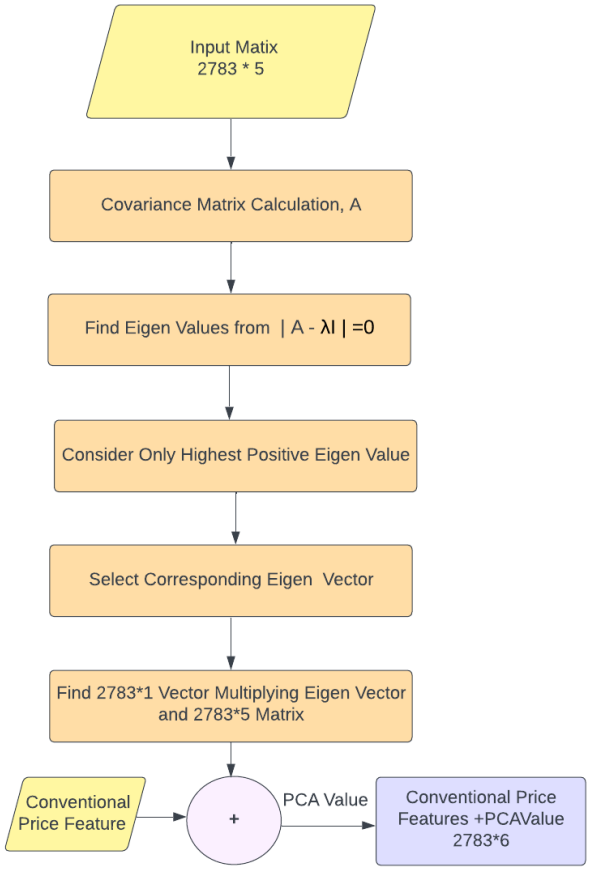

FIGURE 3. Dataset preparation for Chi square testing using PCA value.

Table 2. It also contains another price element derived from the vectors of price elements K1, K2, K3, K4, and K5 using SAM which is known as *'SAM-Price'* shown in Figure 4 that is generated using averaging all the vectors of price element vectors.

Moreover, we experimented with using LDA to derive new price features. This feature has lost all its predictive attributes without applying prediction significance. We experimented with this using Logistic regression that determines the

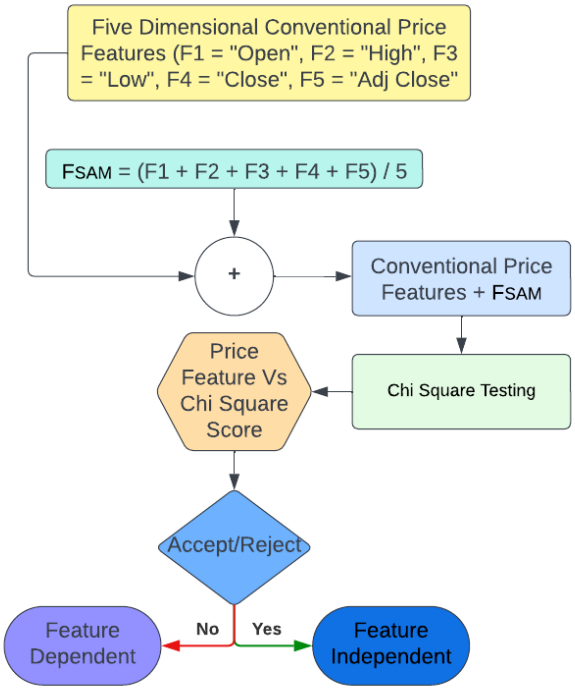

FIGURE 4. Dataset preparation for Chi square testing using SAM value.

TABLE 2. Features name and description.

Feature Name	Feature Symbol	Status
'Open'	K1	Traditional (Conventional)
'High'	K2	Traditional (Conventional)
'Low'	K3	Traditional (Conventional)
'Close'	K4	Traditional (Conventional)
'Adj Close'	K5	Traditional (Conventional)
'PCA-Price'	K(PCA)	Derived (Newly Generated)
'LDA-Price'	K(LDA)	Derived (Newly Generated)
'SAM-Price'	K(SAM)	Derived (Newly Generated)

test value significance. Using this we have determined the feature significance of newly derived price features using test output. The results are discussed below for reference.

We observe that the price feature derived using LDA has the lowest possible significance level compared to both AFD and PFD. Hence, we decided not to use it for price prediction.

The proposed study thus used PCA and statistical averaging methods to create additional price elements. Research predicts prices using three different price data sets.

- **Conventional Price Dataset**: Inclusive price traits will be conventional prices such as *'Open', 'High', Low', 'Adj Close'*, and *'Close'* in this dataset.
- **Price dataset including PCA-Price**: Includes PCA-derived price data with traditional traits as *'Open', 'High', 'Low', 'Adj Close', 'Close'*, and *'PCA-Price'* in this dataset.
- **Price dataset including SAM-Price:**: Includes SAM Price with traditional traits as *'Open', 'High', 'Low', 'Adj Close', 'Close'* and *'SAM-Price'* in this dataset.

C. TRAINING, TESTING AND CROSS VALIDATION

This study focuses on enhancing the accuracy of final price predictions. Our price prediction models aim to forecast missing final prices based on trading policies. Therefore, model robustness analysis is conducted solely on the final set of prices. We indicated using the price data from 01/01/2020 to mid-March for validation purposes. However, the validation accuracies were not explicitly stated due to redundant opportunities. Both 5-fold and 10-fold cross-validation techniques were employed to strengthen the validation model, as detailed in the prediction methods.[11] By defining the predicted price and the actual price, aiming for prediction accuracies of 90% to 110%, the classification-based prediction of stock prices was carried out. An AUC ROC curve was generated based on these parameters, illustrated in Figure 7. According to Hernandez O. Jose while AUC ROC is useful for binary classification tasks, it does not serve an essential purpose in deep learning regression methods, as evidenced by the distinctions in problem nature, relevant evaluation metrics, and interpretability issues associated with regression tasks [84].

1) FEATURE IN-DEPENDENCY USING CHI-SQUARE TEST

In this study, we conducted the chi-square test to evaluate the independence of regular features and derived features. This evaluation is needed to test whether the obtained value characteristics are significant. Our research shows that if every feature possesses a significance value greater than or equal to 10%, then it should be good enough for prediction. In the Chi-Square test, the disparity between the observed value (actual value) and the anticipated amount (predicted value) is calculated. Then, using the Chi-Square distribution table, we found a significant value for the degrees of freedom provided for the typical feature information (5-1) assessment and a confidence interval of 90%. We can now tell if the critical value is greater than the Chi-Square value. If this is the case, our substantial result indicates that independent features are adequate for prediction. PCA was used to convert 5 price features into 1 new feature, with Bi-LSTM providing the highest accuracy. PCA failed to attach to portfolio derivation due to a low chi-square test score. With LDA having a much lower value, it already failed its significance testing. Explained variance is the statistical estimate of deviation attributed to each principal component, with greater explained variances indicating greater importance [85]. Migenda et al. further attributed that PCA eigenvalues have low variance, resulting in inaccurate predictions due to significant information loss. Below is a step-by-step guide to the chi-square significance test used in this research.

[11] https://github.com/hsami180792/Test

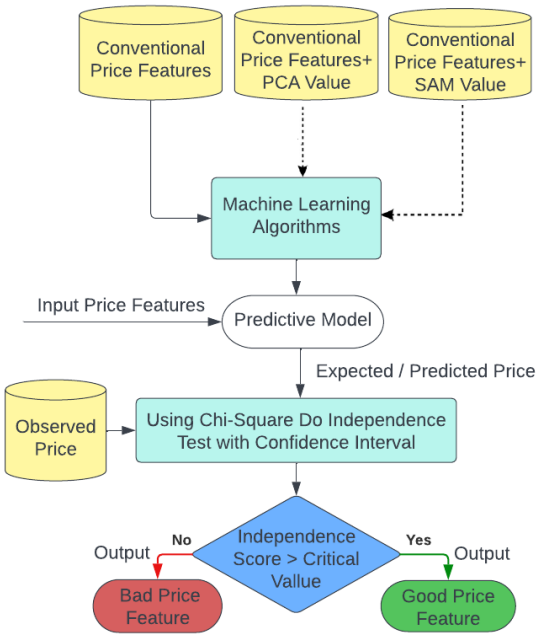

FIGURE 5. **Implementation of Chi-Square test for feature selection.**

Step 1: The training data set will be used to predict values using all supervised learning algorithms.
Step 2: A set of predictive values is generated for all ML processes.
Step 3: In this step, the Chi-Square Test is used to determine the independence score using the difference between the predicted prices and real prices.
Step 4: Critical value and independence score are checked. If the independence score is more than the critical value, then the feature is significant meaning if the predicted price is well represented then it would be accepted as being significant. The following Figure 5 explains the complete process:

PCA is a dimensionality reduction process that has widely been used. As the price features all have very low differences between them in terms of everyday price, we have concluded PCA is an important method to derive new price features. In this process, although during the training process we have dropped the date column we have maintained a strict train test split model to recognize the exact training and testing period required for our research. The date column was dropped during the price derivation process for PCA where all the 5 price features were combined to generate a new price feature.

In our results section, we found that the PCA Price obtained by applying the PCA process to all the regular price features has a significance of 9% which is less than the minimum threshold of 10%. Conversely, the SAM price of average feature-based values has the highest feature independence of 24.4%. Hence, the combined set of average feature-based values and the conventional features should be used to predict stock values.

D. PREDICTING STOCK PRICES

In this investigation, we forecast stock prices using typical price features. We selected to forecast using 5, 6, and 6 price features because Selvin et al. and Wang et al.indicate that predicted stock prices utilizing all price characteristics deliver the best forecasting accuracy. The initial forecast is performed using all five pricing features. For regression analyses based on supervised learning, the proposed research employs the optimum standard deviation of the whole dependent price estimates. Because the 1-dimensional PCA-based feature value does not produce a significant effect, the PCA-based feature value is not to be used for price prediction. The database participated in stock price prediction using 5 and 6 feature criteria, respectively: conventional price features and SAM-price included with conventional price features. In the end, the observed accuracy using the predicted value will determine whether the newly acquired features have improved performance. The following procedure will explain the step-by-step process for estimating stock prices. Also, Figure 6 illustrates the stock price forecasting process that has been introduced in this research.

Step 1: Training is performed using 6 different supervised learning methods that generate trained price models for each price dataset
Step 2: The predictive models for each of the six ML methods have been obtained
Step 3: Price estimation will happen from March 15 to May 29, 2020, using the predictive model.
Step 4: Using the ML rules, generate the outcome of the test period.
Step 5: Compare the estimated and actual values for the test period.
Step 6: Determine the prediction accuracy and classification report for the entire set of values anticipated.
Step 7: Evaluate and finalize the estimates produced by various supervised learning algorithms.
The best ML algorithm for price prediction regardless of stock volatility [86] is selected for portfolio optimization.

E. HYPER-PARAMETER DETAILS

In this research, financial forecasting is carried out using various ML algorithms. Hence, we have made an optimized performance evaluation based on the various optimum performance interfaces of each of these algorithms. We have indulged Optuna a hyper-parameter optimizer to generate the best possible hyper-parameters for the stock price prediction processes [87]. It's observed that in tree-based algorithms Optuna plays an effective role in making accurate predictions [88]. We hence deployed Optuna for all the existing ML models and for neural network-based ML models separately. Optimizing hyperparameters has significantly improved the computation and predictability of the study. Moreover, we have seen the application of Optuna successfully indulged into deep learning-based models which have improved stock price predictions for single price variable. In this experimentation we have brought the novelty of applying

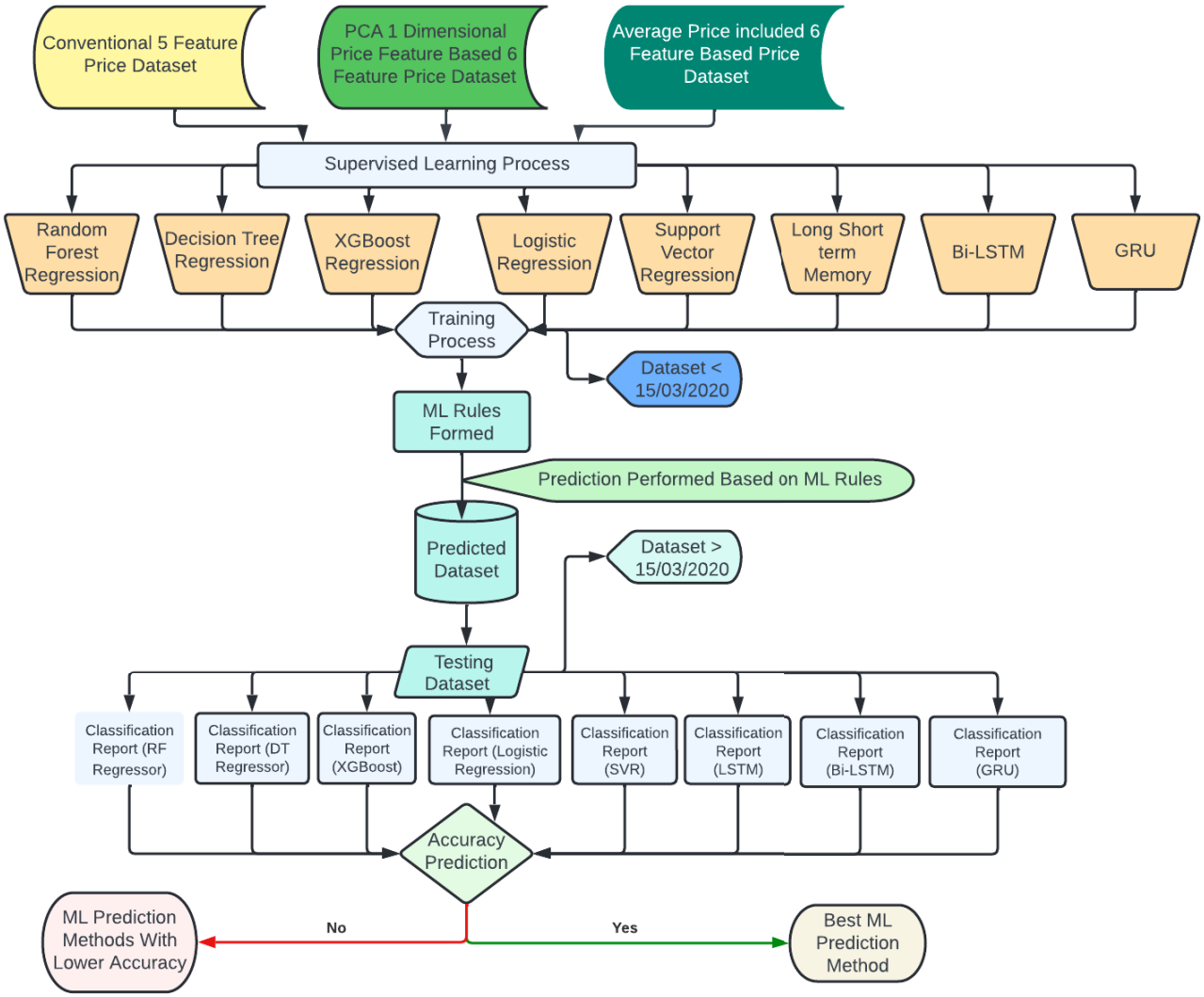

FIGURE 6. ML-based Price Prediction.

multiple price features using optimized hyper-parameters through LSTM for stock price forecasting. Our novelty also dictates the prediction with derived price features that provides better suggestive predictability through chi square optimization [55]. As observed from the literature review, each of these methods is suitable for making effective predictions under various circumstances, but we have not defined the best parameters under which these algorithms can perform well. Hence, we have decided to make an effective selection of prediction parameters that make the best possible prediction. We have found that [38] effective hyper-parameters are very important for the best possible performance. Similarly, we have also seen that accuracy forecasting is only possible when we try the optimized performance parameters for prediction [44]. As our research is dependent on factors such as providing the best possible values for prediction, we are required to determine the best prices for prediction. The following hyperparameter-based findings provide the best possible accuracy found for AFD price dataset using Table 3. We found that the AFD price dataset defines the highest possible accuracy explained in the analysis of the result.

F. CLASSIFICATION REPORT FOR STOCK PRICING

According to Iqbal et al. we have seen that it's easily possible to categorize sentiments through classification processes. Through our research, we also wanted to categorize a forecasted price of very close value from the predicted prices, which were not so accurate. Our research process is tuned to making accurate price predictions. In this research, we experimented by creating a range for predicted prices. We considered that if our predicted prices remained inside that range of 90% to 110% then the prediction is good enough considering that price to be represented in categorization as 1. For forecasted price values other than that, we considered them as 0. We have also proposed categorical values of the

TABLE 3. Hyperparameter details for various ml methods to find the accuracy and r-square score.

ML Methods	Hyperparameter Details	Accuracy (AFD)	Adjusted R^2 (AFD)
SVR	C=20.38, epsilon = 0.001656186, gamma = 'scale'	87.73%	92.3%
DTR	Max_depth=10,max_features='auto', max_samples_split = 7, min_samples_leaf = 1 min_weight_fraction_leaf=0.1,	83.68%	89.50%
LogR	solver='lbfgs', 'fit_intercept' = TRUE,max_iter=1000, RMSE = 1.2747	81.93%	88.71%
RFRegr	n_estimators=250, max_depth=10, , criterion='entropy', min_samples_split=4, min_samples_leaf=11, max_features='auto', bootstrap=True	91.16%	93.40%
Xgboost	eta = 0.49566, gamma = 5.87152, max_depth = 9, min_child_weight = 5,subsample = 0.5472, colsample_bytree = 0.55606	90.17%	85.67%
GRU	Window_size = [1,30], optimzer = "Adam", units = 110, epochs = 627, batch_size = 32 dropout_rate = 0.37611854, learning_rate = 0.002598108, num_layers = 2	95.72%	97.90%
LSTM	optimzer = 'rmsprop', units = 51, epochs = 193, batch_size = 64, dropout_rate = 0.4337508, n_layers = 2, learning_rate = 0.009743564,	96.83%	98.72%
Bi-LSTM	n_nodes = 10, model.add(Dense(10)), model.add((bidirectional(LSTM(dropout = 0.2))), weights = 'random', optimizer = "Adam", learning_rate = 0.0001650593, activation_function = 'ReLU', Momentum = 0.5, Epochs = 60, batch_size = 64	98.14%	99.12%

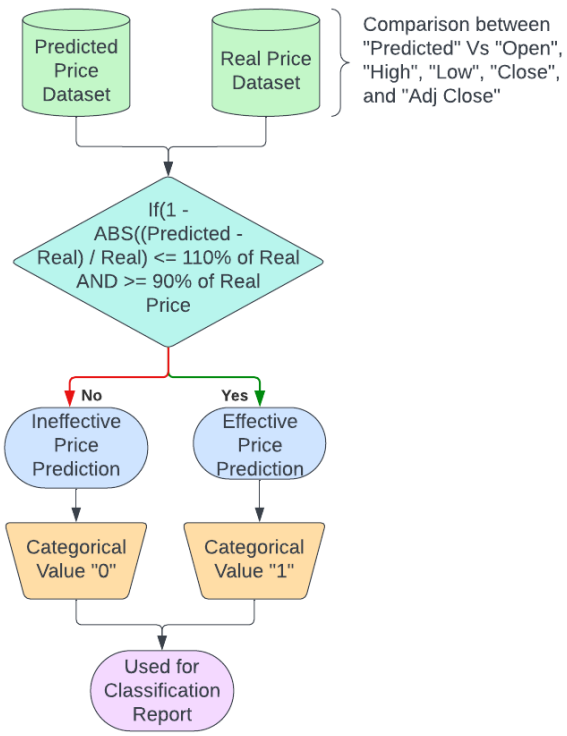

FIGURE 7. Accuracy range of predicted prices for generating classification report.

prediction quality of good predictions and bad predictions by precision, recall, and F1 score. The following figure 7 shows the process of classification report generation.

G. PORTFOLIO OPTIMIZATION

The standard deviation of various stock prices on different dates determines if a stock is volatile or not. A stock is considered volatile if its price varies at random. If the stock price changes with a strong indication of variance, investors will be affected negatively because of the risk of price. Portfolio formulation involves the accumulation of investments that reduce risk and generate more profitable returns [58]. In our research, we created such a portfolio using stock price estimation. These stocks are selected if they show

TABLE 4. Selected stocks for research.

Selected Stocks	Ticker	Beta	EPS	Current Ratio
High Beta Stocks	CELH	2.2	9.78	1.78
	OSTK	4.41	6.34	1.91
	NLS	1.78	4.32	1.62
	TSLA	2.71	2.31	1.73
Medium Beta Stocks	CAT	0.95	11.64	1.24
	VNET	0.98	8.74	1.17
	HD	1.02	2.34	1.31
	TSM	1.01	1.19	1.09
Low Beta Stocks	WYY	0.36	2.31	1.27
	ERIE	0.43	4.65	1.16

an increase in price during the test period. An ideal portfolio will have a Sharpe Ratio value greater than 1. Finally, such optimization is easily produced when using a Python-based library called *PyPortfolioOpt*, which can guarantee the return by reducing the risk of investment while providing the best return for the given risk. Figure 8 illustrates the portfolio optimization using estimated values.

In this research, the investment amount is a small amount of $1,000. This will ensure that the portfolio meets the research objective of small investment. We perform portfolio optimization using approximate values. The algorithm is given below.

Step 1: If the stocks have gained values of their prices at the exercise (Sale) date in terms of predicted price at least by 10% from their purchase price then the stocks should be selected.

Step 2: Using *PyPortfolioOpt* (Python-based library), use a linear program to allocate stocks based on price growth and minimum risk.

Step 3: Evaluate portfolio performance and return on investment (ROI).

The purpose of our research will be to guarantee an effective Sharpe ratio greater than 1 and an ROI of more than 10% per year using the obtained estimate of the average attribute-based price dataset.

IV. RESULTS ANALYSIS

In this study, we have evaluated the results obtained by the proposed methods to inquire about the objectives. Our research motivation is based on stock selection, prediction

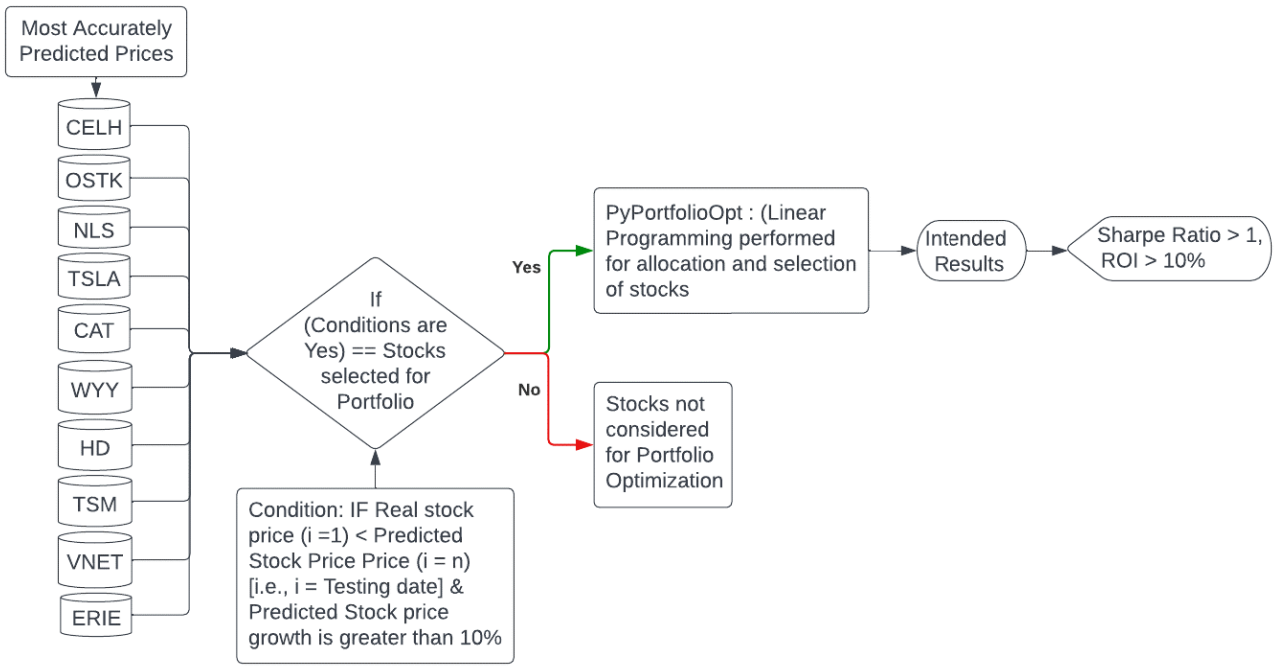

FIGURE 8. Portfolio optimization.

of stock prices, and optimization using predicted stock prices.

A. STOCK SELECTION BY FINANCIAL PARAMETERS

We chose stocks based on financial results and stock beta since Lin et al. proposed that stocks are best chosen based on performance in the market and the financial parameters of each stock [26]. Securities are listed in order of highest volatility in terms of beta [17], [58], [60]. Stocks are created from the selected stock files listed in Table 4.

B. PREPARED PRICE DATASET

This research is formulated using the proposed price dataset. Our primary initiation was to derive three different price datasets including an LDA-based price dataset, a PCA-complied price dataset, and a SAM price-accompanied price dataset. We found that In Table 5, we have shown the existing price features and "SAM-Price" generated using the average statistical method and "PCA-Price" attained from the principal component analysis method, respectively. Due to statistical significance, we have decided to utilize only the SAM price feature for price prediction. The final price datasets that were utilized for this research thus were AFD and CFD.

C. CHI SQUARE TESTING - PRICE IN-DEPENDENCY

In this study, we investigated feature independence using three different price databases. Selvin et al. and Wang et al. found that combining all standard traits effectively predicts price [77], [78]. In their argument, they argued that the future price of a stock is a statistical representation of its past price due to events and financial conditions, so the five price features are not only price features but are also effective information for future price predictions. As each feature indulges required information hence each will have significance for forecasting. Therefore, we considered conducting this study based on the Regular Price Dataset (CFD), the Price Dataset with the PCA Feature (PFD), and the Price Dataset with the Average Feature(AFD). First, we determined whether all the standard features are sufficient to predict the value in terms of feature independence. The chi-square test results for the standard deviation features of the Regular Price Dataset are shown in Figure 9 below.

Evaluating Figure 9, we see that each standard deviation shows greater than 90% independence for a 10% confidence interval for all price features of the regular price dataset. The meaning behind statistical significance is the independence of a specific feature about its future prediction. In addition, based on the ML method, the estimated value obtained with normal features is a great value for predicting the future price. Since PFD and AFD represent two complementary value features obtained by PCA and SAM respectively, observing their importance in terms of value dependence is important. We have not involved a PCA-based price dataset due to feature insignificance for prediction obtained using chi-square tests. Our initial evaluation regarding the significance obtained by LDA-based price features also informed us that the LDA-based price feature could also not be involved in this research.

To understand the regression fitness we have also observed the adjusted R-square score of the price dataset. The results

TABLE 5. A few samples of price dataset including regular and derived price features.

Regular Price Features						Derived Price Features	
Date	*Open*	*High*	*Low*	*Close*	*Adj Close*	*SAM-Price*	*PCA-Price*
7/22/2019	11963.26	11861.68	11998.88	11890.66	11984.27	11939.75	11700.96
7/23/2019	11867.32	11940.15	11871.38	11924.07	11934.31	11907.45	11669.30
7/24/2019	11867.11	11948.42	11884.64	11930.37	11858.15	11897.74	11659.78
7/25/2019	11972.31	11860.97	11878.72	11912.39	11986.11	11922.10	11683.66
7/26/2019	11971.66	11942.78	11873.46	11940.64	11853.96	11916.50	11678.17

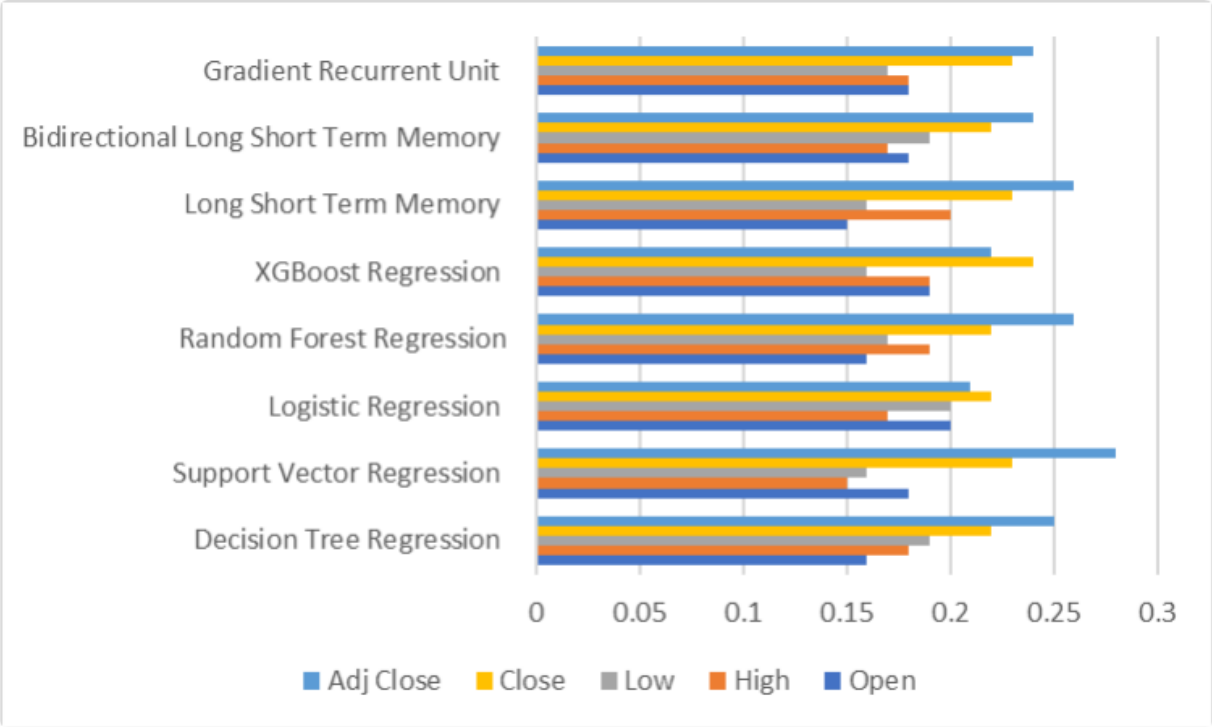

FIGURE 9. Chi square test for conventional Feature-based price dataset.

are shown to explain the efficiency of ML methods through Table 6.

In Figure 9, we also see that the prediction dependency occurs with all price features when it is done in the training value database using the ML algorithm. We can see that the feature value derived from PCA has the lowest level of value dependence, with a value less than 10%, indicating a low value for price prediction. Similarly, PFD has been observed several times to show high variance dependence, so the price database is ineffective for price prediction. This proves that we should not use PFD for price prediction in this study. Therefore, our research proceeds by using CFD and AFD as two effective price databases to be used for stock price prediction using ML techniques. The results of using the proposed value set are shown in Figure 10.

Our study also tested chi-square with a 90% confidence interval. We see that the PCA-based price feature shows a 9% dependence level and the average feature shows a 24.4% independence level, thus making this new feature the most effective in predicting prices.

D. OVER-FITNESS ANALYSIS

In this research, we have focused on understanding if the proposed price features perform well in comparison to the validation and test dataset. We thus have ensured a validation testing of a dataset containing price dataset values from 1st January 2020 till 15th March 2020 of the training dataset to evaluate the model's forecasting accuracy. The following table 7 exhibits an over-fitness analysis of the proposed price features.

We observe that the accuracy achieved for the validation dataset remains the same compared to the accuracy values found in both CFD and AFD-based price dataset table 7. Hence we can consider that the dataset exhibits no over-fitting or that the test dataset and validation dataset exhibit the same accuracy.

TABLE 6. Adjusted r-square values of different ML algorithms for AFD and CFD.

Stocks	SVR		LogR		RFRegr		DTR		XGBoost		LSTM		Bi-LSTM		GRU	
	AFD	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*
CELH	0.89	0.88	0.9	0.86	0.86	0.83	0.92	0.87	0.93	0.87	0.95	0.88	0.98	0.91	0.95	0.88
OSTK	0.93	0.87	0.86	0.82	0.88	0.86	0.91	0.83	0.88	0.82	0.92	0.91	0.98	0.92	0.95	0.91
NLS	0.91	0.87	0.87	0.83	0.89	0.87	0.88	0.88	0.89	0.83	0.9	0.89	0.98	0.95	0.93	0.91
TSLA	0.89	0.82	0.87	0.83	0.89	0.89	0.9	0.84	0.88	0.86	0.91	0.87	0.97	0.92	0.96	0.92
CAT	0.91	0.89	0.88	0.87	0.9	0.84	0.88	0.85	0.91	0.87	0.94	0.93	0.96	0.93	0.92	0.89
DE	0.94	0.89	0.92	0.9	0.9	0.86	0.89	0.86	0.87	0.87	0.98	0.87	0.97	0.92	0.91	0.88
HD	0.9	0.84	0.87	0.82	0.87	0.82	0.87	0.85	0.89	0.88	0.9	0.9	0.95	0.93	0.9	0.87
TSM	0.91	0.84	0.92	0.88	0.94	0.87	0.91	0.82	0.87	0.84	0.89	0.86	0.92	0.89	0.9	0.87
VNET	0.9	0.85	0.88	0.84	0.89	0.83	0.86	0.83	0.89	0.85	0.91	0.88	0.91	0.92	0.89	0.84
ERIE	0.93	0.92	0.88	0.87	0.88	0.86	0.88	0.88	0.89	0.86	0.91	0.9	0.95	0.91	0.9	0.88

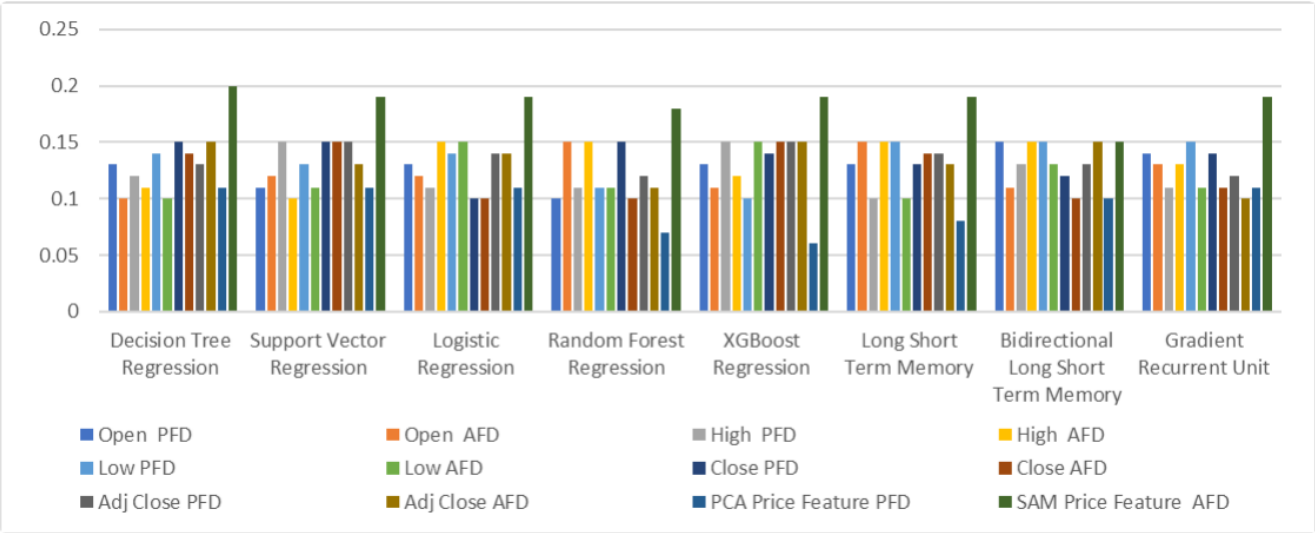

FIGURE 10. Chi square testing for proposed price dataset.

E. STOCK PRICE FORECASTING

To understand the predictive ability of the ML algorithm, we evaluated the predicted values against the classification report. The results are shown in Figure 11. In the case of CFD, each classification report has dimension parameters slightly lower than AFD.

Figure 11 shows that our proposed estimation feature improves the estimation performance. It also shows that in the case of CFD and AFD, Bi-LSTM makes the most efficient prognosis. Therefore, it is concluded that Bi-LSTM is still the best value prediction ML algorithm among the various ML methods. In Table 8, we show evidence of prediction accuracy achieved by each of the different ML methods used in this study for all stocks. The accuracy is derived using the difference between the predicted price and the real price which is divided by the actual price of the stock, and then subtracting the result from one, as illustrated in equation 1. This result defines how much difference the predicted prices have in comparison to the real stock prices. Mostly this process shows how accurate our stock forecasting methods are concerning both highly volatile stocks and non-volatile stocks.

$$Accuracy = 1 - \frac{|\text{Observed Price - Actual Price}|}{\text{Actual Price}} \quad (1)$$

We can see that nine out of ten stocks have higher accuracy using the proposed value set (AFD) and only one has higher accuracy using the standard feature-based value set (CFD). There are six (6) volatile stocks in our database. While the nine (9) price predictions are more accurate using the recommended value of our proposed price dataset, hence we can recommend that the average price feature thus helps make better predictions for both volatile and non-volatile stocks. Figure 12 thus illustrates the accuracy and mean error (MAE)

TABLE 7. Overfitting result analysis using validation dataset.

Overfitness	SVR		DTR		LogR		RFRegr		Xgboost		GRU		LSTM		Bi-LSTM	
	CFD	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*	*CFD*	*AFD*
CELH	0.88	0.86	0.89	0.88	0.89	0.89	0.88	0.89	0.88	0.89	0.89	0.86	0.89	0.89	0.91	0.93
OSTK	0.82	0.94	0.86	0.89	0.85	0.87	0.88	0.94	0.89	0.95	0.86	0.94	0.89	0.95	0.91	0.93
NLS	0.87	0.91	0.89	0.93	0.86	0.94	0.91	0.94	0.93	0.96	0.88	0.92	0.89	0.92	0.92	0.97
TSLA	0.87	0.93	0.87	0.91	0.88	0.93	0.91	0.94	0.88	0.92	0.92	0.95	0.96	0.96	0.91	0.93
CAT	0.93	0.94	0.89	0.95	0.87	0.92	0.92	0.95	0.94	0.92	0.93	0.93	0.89	0.91	0.91	0.96
WYY	0.89	0.89	0.84	0.89	0.87	0.89	0.83	0.77	0.86	0.89	0.89	0.91	0.87	0.93	0.91	0.96
HD	0.91	0.94	0.89	0.93	0.94	0.93	0.89	0.89	0.88	0.89	0.84	0.87	0.88	0.92	0.91	0.94
TSM	0.91	0.94	0.92	0.91	0.89	0.88	0.89	0.92	0.87	0.94	0.95	0.93	0.91	0.92	0.89	0.93
VNET	0.94	0.87	0.89	0.88	0.89	0.85	0.89	0.88	0.83	0.89	0.87	0.91	0.88	0.92	0.89	0.93
ERIE	0.89	0.91	0.88	0.91	0.85	0.88	0.87	0.89	0.88	0.92	0.87	0.89	0.89	0.91	0.88	0.97

TABLE 8. Accuracy of various ML methods for AFD and CFD.

ML Methods	Dataset	CELH	OSTK	NLS	TSLA	CAT	WYY	HD	TSM	VNET	ERIE	Average
SVR	AFD	81.6	86.4	88.9	86.8	85.9	81.5	84.7	83.6	80.3	78.4	85.47
	CFD	79.6	78.3	81.2	83.2	81.3	86.1	85.9	83.8	85.9	87.5	87.42
DTR	AFD	84.5	89.3	84.3	85.6	86.4	86.6	81.8	80.7	80.78	89.2	86.68
	CFD	79.1	81.8	82.38	83.26	81.79	81.2	84.7	78.91	79.4	80.28	81.18
RFRegr	AFD	83.2	84.6	86.2	84.9	89.2	89	84.8	80.72	83.48	89.7	87.16
	CFD	83.4	86.8	89.6	82.7	82.8	82.1	85.4	82.78	88.1	81.1	85.4
Xgboost	AFD	82.4	83.18	81.35	81.9	84.56	81.4	86.8	89.1	89.7	87.6	87.08
	CFD	78.1	76.7	79.3	79.3	78.5	78.37	79.4	77.7	78.2	79.2	78.84
LogR	AFD	83.1	83.8	81.2	84.6	84.4	83.8	81.7	85.5	84.7	85.5	85.93
	CFD	77.3	79.5	78.4	76.8	77.1	78.1	81.4	79.3	78.3	78.6	77.48
LSTM	AFD	94.6	92.7	94.7	93.6	92.2	94.9	90.9	92.7	92.6	95.5	93.28
	CFD	91.5	90.4	91.1	89.3	91.4	90.4	92.6	89.4	90.4	90.72	91.37
GRU	AFD	90.72	91.12	90.9	91.5	93.6	92.2	94.3	95.5	94.1	91.9	93.02
	CFD	89.8	88	89.13	90.38	90.12	93.8	89.4	89.6	87.92	88.71	89.56
Bi-LSTM	AFD	97.6	96.7	98.5	98.1	97.3	98.7	97.2	98.6	97.6	96.8	97.74
	CFD	94.3	91.7	94.4	92.1	92.5	93.2	94.3	95.2	93.37	93.5	92.55

for the estimate obtained using the proposed set of values (AFD).

F. PORTFOLIO OPTIMIZATION

In this research, Bi-LSTM produces the best predictions using CFD and AFD datasets. For this study, we consider stocks that have risen at least by a factor of 10% during the test period. In addition, the ML validation method is adopted for price prediction because about 60% of the stocks considered in our portfolio have high volatility and the trial period is not suitable for financial forecasting due to the pandemic factor. Table 9 describes the stocks to consider for portfolio optimization using a 10% or greater positive return.

Our portfolio contributed a requisite margin of USD 1000 as the contribution margin for the portfolio. We considered the 15th, March 2020 date as the purchase date for the stocks. Based on our research contribution we wanted to observe a market portfolio performance based on the most volatile periodic return. We considered 30th, June 2020 as the portfolio liquidity date. We applied PyPortfolioOpt as the optimization guideline that enables the minimum risk and maximum return benchmark. According to this process, we found that "NLS", "CELH", "VNET" and "TSLA" played an essential role in portfolio optimization. We expected a yearly annual return of 1085% approximately for this portfolio investment. We found that the portfolio suggested 17 stocks of NLS, 124 stocks of CELH, 2 stocks of TSLA, and 28 stocks of VNET by the PyPortfolioOpt-based optimization.

TABLE 9. Forecasted stock price using predicted Vs. Real price.

Stock Tickers	Real Price (03/15/2020)	Real Price (06/30/2020)	Predicted Price (06/30/2020)	Growth Percentage
CELH	4.44	5.29	5.31	0.19144144
OSTK	1.69	2.76	2.77	0.63313609
NLS	1.22	2.73	2.76	1.23770492
TSLA	554.65	659.47	657.48	0.18898404
CAT	231.45	229.18	230.39	-0.0098077
WYY	2.4	3.4	3.41	0.41666667
HD	109.11	111.64	112.16	0.02318761
TSM	102.35	102.97	103.17	0.00605765
VNET	4.96	5.39	5.41	0.08669355
ERIE	34.55	32.91	33.04	-0.0474674

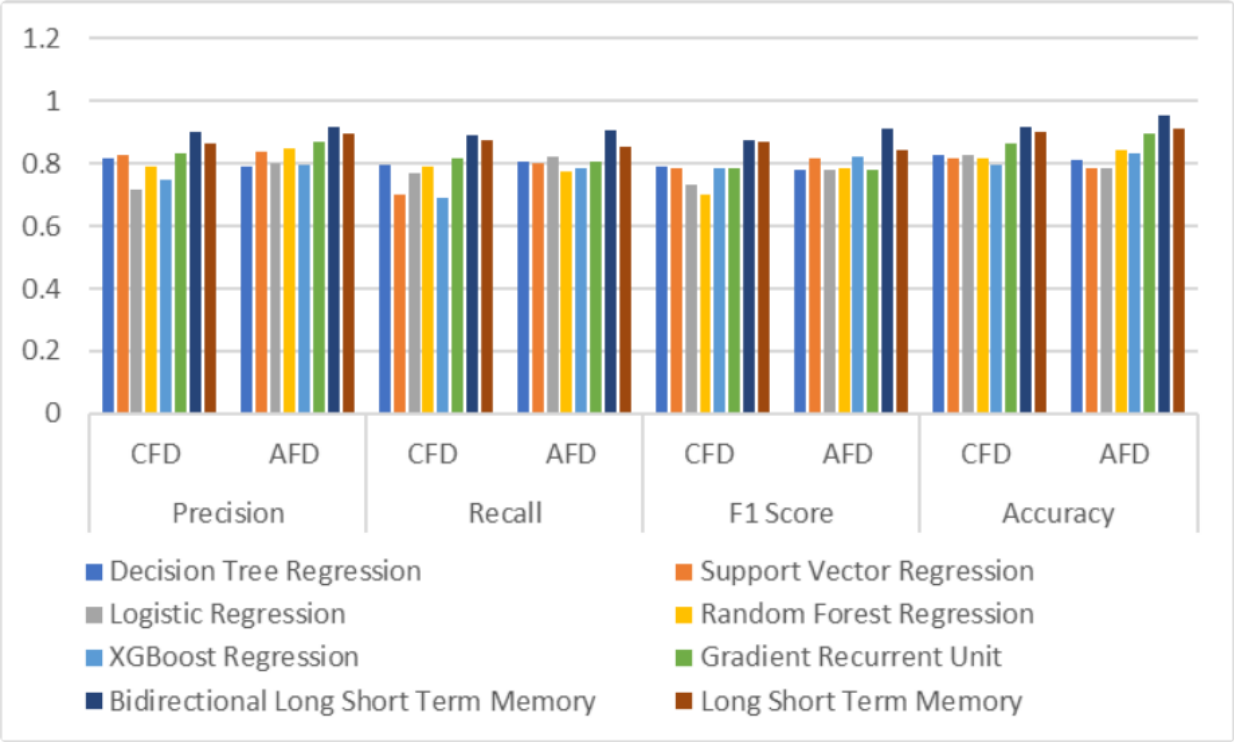

FIGURE 11. Classification report for conventional & proposed price dataset.

PyportfolioOpt is a Python-based optimization library that assigns stocks with the lowest risk and highest return. The portfolio efficiency of two portfolios established with expected values is used to quantify portfolio correctness.

- Portfolio with Predicted Price Obtained by Conventional Price Features (CFD).
- Portfolio with Predicted Price Obtained by Average Feature-based Price Dataset (AFD).

G. COMPARISON: PROPOSED MODEL VS EXISTING MODEL

In this research, our goal was to optimize portfolio performance using multi-variable feature-based price forecasting methods. To achieve better forecasting accuracy we have shown the chi-square test value of selected additional price features, the Over fitness Analysis of AFD price dataset, and also the prediction accuracy of the AFD-based price dataset which outperformed the CFD dataset in all aspects through predetermined ML forecasting techniques. In Table 10, the research outcome is discussed and explained in comparison to existing model applications and novel models.

Finally, using the forecasted prices the portfolio optimization has ensured a better return on investment and a better Sharpe ratio rather than CFD based portfolio. Hence we can presumably confer that our research for novel price dataset development has been significant for price prediction and portfolio performance optimization.

A new feature offered with portfolio price averaging improves stock forecasting. Similarly, by including the average value of all real values as an attribute in the price database, the portfolio optimization is enhanced.

Table 11 shows that ROI and other benchmarks report about the portfolio productivity. Portfolios constructed using our recommended value sets have been observed to generate more efficient returns in terms of investment returns.

Therefore, we can conclude that by including the mean value as an additional value, the feature helps to predict both prices and the accuracy of the portfolio with more efficiency.

- Generate better price estimates when making price forecasts using multiple features. (Using Bi-LSTM, the average feature-based set of attributes generates 98. 14% precision, while the standard set of price features obtains 92. 55% precision).
- A portfolio obtained using the price estimate from the proposed price base has an ROI of 55.589% compared to 46.76%. The return of the portfolio constructed using the actual value is 138.45%. The predictions obtained using AFD seem to have better performance accuracy than CFD. This demonstrates that the prices in our proposed pricing database are superior to those that guarantee better return potential and price forecasts.

H. PREDICTION & PORTFOLIO OPTIMIZATION: BI-LSTM APPLICATION

Our research found that Bi-LSTM has accumulated the highest possible price precision using the AFD price dataset. In the Bi-LSTM method, we have identified the best possible hyperparameters for our price prediction. Using Optuna

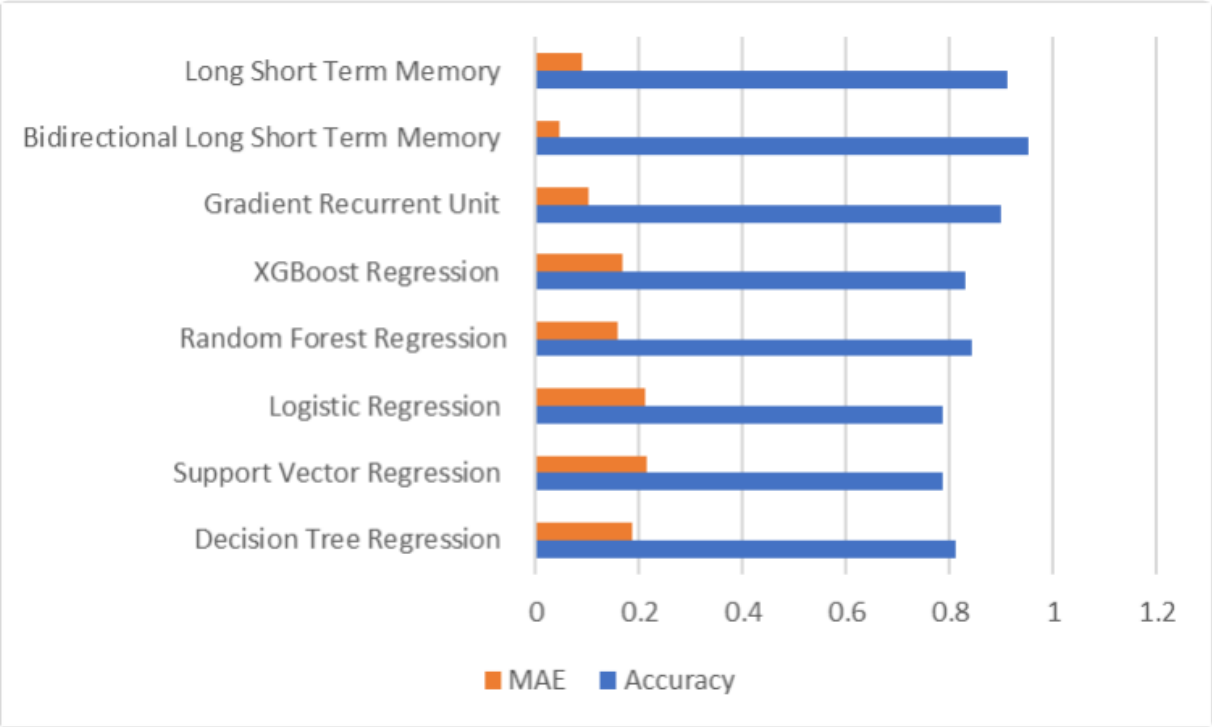

FIGURE 12. Accuracy and MAE by different ML Methods - For AFD.

the hyperparameters were derived after 1000 repetitions. The inference is that the Bi-LSTM method with better hyperparameters would attain better predicted prices, which as a result would obtain better portfolio performance because better predicted prices would be able to provide more accuracy in comparison to real portfolio performance. Bidirectional LSTM played a crucial role in making the most accurate predictions. Although neural networks and CNN-based predictions have remained successful for stock price forecasting due to effective learning structure, bidirectional LSTM with its effective hyperparameters has played a crucial role in effectively consuming the best accuracy. Using Optuna we have indulged in developing multi-optimized hyperparameters for stock price predictions. The hyperparameter optimizations are checked with approach for correction and optimizing the performance for accuracy improvement through a set of parameters discussed in the application of various input parameters for BiLSTM. It resulted in generating various sample corrections and finally produced the results of 'random' weight allocations, with Adam Optimizer, a learning rate of 0.0001650593, ReLU activation function, a momentum of 0.5, and Epochs highest at 60 and a batch size of 64. This resulted in producing an accuracy rate of 98.14% with an adjusted r^s *coreof* 99.12%

V. DISCUSSION & LIMITATION

The main objective of this study is to employ supervised learning techniques to predict stock prices based on various features related to the stock market.

A. DISCUSSION

This investigation was carried out to deal with various issues associated with price prediction and effective ROI generation. Hence various issues could be discussed further below:

- Using PCA and SAM approaches, two new value characteristics are introduced. The chi-square test demonstrates that there is considerable dependence on the attributes' features. The PCA-based estimate had the lowest specificity at 9% dependency, while the recommended mean estimate had the highest dependency at 24.4%. Using a set of SAM-based price features along with the AFD dataset to anticipate prices results in good forecasting accuracy.
- It is also observed that the new value of "SAM-Price" attuned with the AFD price dataset has the lowest standard deviation compared to the normal price characteristics. We have found the SAM price to attain the highest chi-square score. Therefore, we can say that the average feature-based property can make better stock predictions than only standard feature-based properties.
- Price prediction accuracy is about 4% higher using the proposed price features of AFD and about 8% better for the return on income obtained using the allocation suggestion defined by PyportfolioOpt.
- LDA-based price features are found to have very low significance. In comparison to the PCA-based price features, we found that they have much lower price independence rather than PCA-based features. The Chi-Square test performed provides much lower

TABLE 10. A comparison table of the existing state-of-the-art methods with proposed model.

Existing Litterateur	Method	Accuracy	Features	Feature Names
Y. Yang (2021)	XGBoost	71.23%	1	Derived feature from "Adj Close"
Gumelar et al. (2020)	XGBoost	79.12%	1	Close
Gong & Sun. (2009)	LogR	87.11%	5	Open, High, Low, Close, Adj Close
Ali et al. (2018)	LogR	89.77%	1	Close
Hindrayani et al. (2020)	DTR	74.32%	1	Adj Close
Vijh et al. (2020)	RFR	70.39%	5	Open, High, Low, Close, Adj Close
Henrique et al. (2018)	SVR	76.64%	1	Close
Kao et al. (2013)	SVR	71.12%	1	Adj Close
Moghar & Hamiche (2020)	LSTM	81.22%	1	Adj Close
Wang et al. (2018)	LSTM	97.66%	5	Open, High, Low, Close, Adj Close
Research Experimentation	**Method**	**Accuracy**	**Features**	**Feature Names (Only AFD Price Dataset)**
Dataset (CFD, AFD)	XGBoost	87.08%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	SVR	87.42%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	LogR	85.93%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	DTR	81.18%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	RFR	85.40%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	LSTM	93.28%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	Bi-LSTM	93.02%	6	Open, High, Low, Close, Adj Close, SAM Price
Dataset (CFD, AFD)	GRU	98.14%	6	Open, High, Low, Close, Adj Close, SAM Price

TABLE 11. Compendium of proposed research.

Portfolio Parameters	Result	Benchmark
Predicted Portfolio	55.589%	Real Portfolio will have 100% Accuracy
Two Months Gain	145.475%	Yearly S&P return is 10.5%
Expected Annual Return	355.64%	Highest annual savings account interest is 7.8%
Annual Volatility	45.8%	S&P Yearly Risk is 22.35%
Portfolio Accuracy	71.48%	Effectiveness must be more than 50%
Approximate Gain	689.87	Profit must be more than zero (0)
Actual Gain	1454.75	Profit must be more than zero (0)
Sharpe Ratio	77.62	needs to be more than one (1)

significance for LDA-based price features[12] rather than PCA-based features. Similarly, we find the LIME and SHAP analysis of the LDA-based price to be very low, hence that price is ignored for price prediction.

- Adverdently we found that LIME and SHAP analysis denoted the highest degree of significance for forecasting through finding the SAM price feature. Hence we can consider that the chi-square test be directly correlated with LIME and SHAP analysis findings.[13] The AFD price dataset which includes the SAM price feature thus generated forecasted prices with the highest accuracy. Also, there may be a discussion question that ascertains the significance of the newly found price feature as that is not a real price. But as the price is the average of all prices hence this price itself contained the highest significance in comparison to other conventional price features. This means our newly generated attribute contains important predictive information for prediction.

B. LIMITATIONS OF RESEARCH

This study has a few constraints that could not be addressed due to the research method of this experimentation. We did not initiate price forecasting using investor behavioral analysis as an important contributor. Our sole analysis only contained regression based on training and testing performance. Hence more effective investor analysis would give better price forecasting accuracy in future research. The following limitations of the study are mentioned below:

- The investigation solely examines conventional machine learning (ML) algorithms in the context of stock pricing procedures. It only focuses on predicted prices and employment of those prices in the portfolio for higher than market standard return. However, this research did explore the opportunity of prediction using only the obtained price features. The impact of news impact, major commodity price movement, and external environmental factors on price forecasting has been ignored.
- Our study emphasizes achieving accuracy and profit generation as effective outcomes rather than prioritizing the computational speed required to produce results. Hence we were not able to make forecasts on very short-duration changes that could be more beneficial for the investors.
- This research does not delve into a novel approach using supervised regression that can facilitate more accurate short-term predictions with a reduced margin of error and a smaller training dataset.
- There is potential for further exploration by employing stocks with lower standard deviations and more favourable financial conditions.

VI. CONCLUSION

This research found that combining the statistical averaging method (SAM) feature with the existing stock price feature improves the accuracy of predictions when using Bi-LSTM. In the case of Bi-LSTM, a set of values with a lower deviation helps the neural network gain more information about predicting future values. As a result, we can make

[12] https://github.com/hsami180792/Test
[13] https://github.com/hsami180792/Test

recommendations based on a complete examination of feature-based price forecasting and portfolio optimization, which has made the forecasting process efficient and effective. The summarized outcomes of our proposed approach are as follows.

- PCA-based price traits have 9% independence, hence they are not particularly useful for predicting stock prices.
- The average price demonstrates 24.4% independence, making it the most essential pricing attribute for predicting our stock price.
- The multi-price feature, which includes the average price function, has the greatest accuracy in forecasting stock prices, with a 98.14% accuracy.
- A two-month portfolio derived using the estimated value of the average feature-based database showed an ROI of 55.589% predicted return.

The above evidence proves that the AFD price dataset can improve the price prediction of a set of prices obtained using the additive price feature. The research aims to reach an annualized return of more than 10% (S&P 500 Annual Return) and a savings account return of over 8% (Best Savings Account) utilizing standard features and recommended stock price values. As a result of our research, modest investors can dream of earning secondary income in the market through ML-based predictions.

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